DIVYA YADAV

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SUMMARY

● Risk and analytics professional with 8+ years in regulated financial services (Bank of Baroda) and a Master of Management in Risk and Analytics from NC State (4.0 GPA).

● Applies Python, SQL, R, and statistical analysis to business problems, building and evaluating machine learning models (gradient boosting, random forest, K-Means) and Monte Carlo decision models. Trained in applied AI and data science through MIT's Applied AI and Data Science Program.

● Brings credit risk, regulatory, and compliance context to translating business questions into quantitative analyses, evaluation metrics, and decision-ready recommendations for cross-functional stakeholders.

TECHNICAL SKILLS

●**Programming & Querying:** Python (pandas, NumPy, scikit-learn), SQL (joins, aggregations, analytical queries), R, SAS

●**Machine Learning & Statistics:** Supervised and unsupervised learning, regression, logistic regression, decision trees, random forest, gradient boosting, K-Means clustering, cross-validation, feature engineering, feature importance, hyperparameter tuning, model evaluation (AUC, R^2), statistical modeling

●**Analytics & Decision Science:** Predictive analytics, forecasting, Monte Carlo simulation, decision analysis (DPL), exploratory data analysis, data preprocessing, financial modeling

●**Data Engineering & BI:** Snowflake, ETL pipelines, Alteryx, data warehouse concepts, database querying, Power BI, Tableau, JMP, Excel

●**Risk & Domain:** Credit risk analysis, enterprise risk management, portfolio analysis, regulatory and compliance environments, risk communication

PROFESSIONAL EXPERIENCE

Bank of Baroda — Business Associate, Banking & Financial Services, *India* | *Jul 2015 – Feb 2026*

●Analyzed borrower financial statements, credit data, and repayment capacity to support risk-informed lending decisions in a regulated, policy-driven banking environment.

●Evaluated loan applications against credit policy and regulatory requirements, documenting the risk rationale behind each recommendation to support approval, audit, and compliance review.

●Monitored loan portfolio and borrower account data to flag emerging credit risk early and inform portfolio management and follow-up actions.

●Reviewed banking operations and business processes to identify bottlenecks and control gaps, recommending process changes to improve turnaround and consistency.

●Translated financial and risk analysis into clear recommendations for branch leadership, credit, and operations teams, supporting cross-functional, data-driven decisions.

●Advised customers and borrowers on financial products and obligations, balancing customer needs with the bank's risk appetite and regulatory obligations.

SELECTED DATA SCIENCE & ANALYTICS PROJECTS

Loan Default Prediction — Credit Risk Capstone | *MIT Applied AI and Data Science Program*

●Built a leakage-safe scikit-learn pipeline on 5,960 home-equity loan applications (20% default rate), tuning logistic regression, decision tree, random forest, and XGBoost for recall with class weighting and a 5-fold cross-validated search.

●Selected tuned XGBoost (test recall 0.80, precision 0.76, ROC-AUC 0.95) and recommended a 0.30–0.40 decision threshold that caught 32 more defaulters on the test set, paired with an interpretable logistic regression model for explainable decisions.

●Found that applicants with a missing debt-to-income ratio defaulted at 62% vs. 9%; audited my own pipeline to fix a threshold-analysis bug and documented calibration, selection bias, and fair-lending limitations.

Telehealth Expansion Prioritization — Predictive Modeling & Clustering | *NC State, MBA 551*

●Built a random forest regression model on county-level data for 3,143 U.S. counties (R^2 = 0.81) to support telehealth expansion planning.

●Segmented counties with K-Means clustering and combined model output with segment profiles to identify 477 priority counties for telehealth expansion, presented as a consulting-style recommendation.

Profit-Driven Classification & Cost-Benefit Analysis | *NC State, Predictive Analytics*

●Built and compared logistic regression, random forest, and gradient boosting classifiers in Python (scikit-learn) on 100,000 records with a 6.5% positive rate, selecting gradient boosting (test AUC 0.63) using ROC and lift analysis instead of accuracy.

●Converted predicted probabilities into an expected-profit targeting rule, nearly tripling test-set net profit versus contacting everyone ($7,493 vs. $2,649).

House Price Prediction — Linear Regression | MIT Applied AI and Data Science Program

●Built an OLS regression model in Python (statsmodels) on 506 Boston-area census tracts, with IQR capping of outliers and a log-transformed target to meet regression assumptions.

●Refined the model with backward elimination, VIF multicollinearity checks, and residual diagnostics; the final 9-predictor model reached a test R^2 = 0.83 and a MAPE of 12.5%, with a 5-fold cross-validated R^2 of 0.77.

Food Delivery Demand & Operations Analysis | MIT Applied AI and Data Science Program

●Analyzed 1,898 orders across 178 restaurants in Python (pandas, seaborn) to profile demand, delivery performance, ratings, and revenue under a tiered commission model.

●Found that weekday deliveries ran about 6 minutes slower despite lower volume and that orders over $20 drove about 60% of platform revenue; delivered a business report with operational and pricing recommendations.

KidTech Production Capacity Decision — Decision Analysis | NC State, MBA 545

●Built an Excel cash-flow model and a DPL decision tree to compare four capacity strategies, using tornado sensitivity analysis to identify demand growth as the key uncertainty.

●Ran a probabilistic evaluation with Extended Swanson-Megill approximations and cumulative risk profiles, recommending a royalty partnership with the highest expected NPV ($69.1M).

Cloud Data Pipeline & Data Engineering | *NC State, MBA 552*

●Built an ETL pipeline in Snowflake (team project) to ingest, transform, and query data with SQL for downstream analytics; built a separate Alteryx/Python data-preparation workflow.

Asset Pricing Regression Analysis | *NC State, Graduate Certificate in Finance*

●Estimated CAPM and Fama-French factor regressions in SAS on 40 years of CRSP return data to test risk-return relationships.

EDUCATION

North Carolina State University — Poole College of Management | Raleigh, NC

Master of Management in Risk and Analytics | GPA: 4.0 | June 2026

●**Relevant coursework:** Predictive Analytics, Data Engineering, Decision Making Under Uncertainty, Forecasting & Monte Carlo Simulation, Strategic Risk Analysis, Enterprise Risk Management, Data Analytics

Graduate Certificate in Finance | GPA: 4.0 | July 2025

●**Relevant coursework:** Advanced Corporate Finance, Investment Theory and Practice, Financial Management, Quantitative Analysis

University of Kota — Master of Commerce (M.Com.) | 2013

Bachelor of Commerce (B.Com.) | 2011

CERTIFICATIONS & HONORS

●**MIT—Applied** AI and Data Science Program (Professional Certificate)—2026

●SQL for Data Science—University of California, Davis—2025

●Database Management Essentials; Data Warehouse Concepts & Design—University of Colorado 2025

●Google Data Analytics Professional Certificate — 2026

●Beta Gamma Sigma International Business Honor Society — 2026